

Literature-implied answer to Remark 5.2 of arXiv:1302.3372

The strong-dual and inductive-limit Wiener topologies coincide

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Status. `literature_implied_answer_full_scope`. The answer is affirmative under all hypotheses stated in the source remark. This is not a new theorem: it follows from a 1999 result of Diaz and Domanski after a direct identification.

Question in the source

Alpay and Salomon assume that

$$\Phi_1 \supseteq \Phi_2 \supseteq \cdots, \quad E := \bigcap_{p \geq 1} \Phi_p$$

is a reflexive Frechet space, with each Φ_p a reflexive Banach space. They put

$$\Psi_p = c_0(\mathbb{Z}; \Phi_p), \quad \Psi'_p = \ell_1(\mathbb{Z}; \Phi'_p),$$

identify $\bigcup_p \Psi'_p$ with the associated Wiener algebra, and ask in Remark 5.2 when its inductive-limit topology coincides with its topology as the strong dual of $\bigcap_p \Psi_p$ [1, Remark 5.2].

Identification and answer

Let $K = \mathbb{Z} \cup \{\infty\}$ be the one-point compactification of the discrete space $\mathbb{Z}$. With the natural Frechet seminorms,

$$\bigcap_{p \geq 1} c_0(\mathbb{Z}; \Phi_p) = c_0(\mathbb{Z}; E).$$

Indeed, membership on either side says exactly that $\sup_n \|x_n\|_p < \infty$ and $\|x_n\|_p \rightarrow 0$ as $|n| \rightarrow \infty$, for every p .

Moreover,

$$c_0(\mathbb{Z}; E) = \{f \in C(K, E) : f(\infty) = 0\}$$

is complemented in $C(K, E)$. A continuous projection is

$$(Pf)(n) = f(n) - f(\infty), \quad (Pf)(\infty) = 0.$$

Diaz and Domanski proved that for every reflexive Frechet space E and every compact Hausdorff K , the space $C(K, E)$ is distinguished [2]; the theorem is also recorded explicitly in the Bierstedt–Bonet survey [3, p. 183]. Distinguishedness passes to complemented Frechet subspaces, so $c_0(\mathbb{Z}; E)$ is distinguished.

For a Frechet space F , distinguishedness is equivalent to the equality of the strong-dual topology $\beta(F', F)$ and the inductive topology on F' obtained from the duals of its local Banach spaces [3, Sec. 3(a)]. Here those local Banach spaces are the Ψ_p , whose duals are $\ell_1(\mathbb{Z}; \Phi'_p)$. Consequently,

$$\beta\left(\left(\bigcap_p \Psi_p\right)', \bigcap_p \Psi_p\right) = \varinjlim_p \ell_1(\mathbb{Z}; \Phi'_p).$$

Thus the two topologies in Remark 5.2 always coincide under the assumptions of that remark.

Why this is literature-implied rather than new

The supporting result predates the 2013 source paper, so its authors could not explicitly identify their theorem as an answer to Remark 5.2. The specialization through the one-point compactification and the complemented zero-limit subspace is the present identification. Accordingly this packet records a full-scope literature-implied answer, not an original solution.

Search record and review recommendation

The bounded search used the exact source sentence and the terms *distinguished Frechet*, $C(K, E)$, *injective tensor product*, and the source title. It located the official abstract of [2] and the survey statement [3, p. 183]; no later paper explicitly naming the source remark was found. Human review should check only the standard stability of distinguishedness under complemented subspaces and the local Banach-space identification above.

References

- [1] D. Alpay and G. Salomon, *On algebras which are inductive limits of Banach spaces*, Integral Equations Operator Theory 77 (2013), 63–78; arXiv:1302.3372.
- [2] J. C. Diaz and P. Domanski, *On the injective tensor product of distinguished Frechet spaces*, Math. Nachr. 198 (1999), 41–50, doi:10.1002/mana.19991980104.
- [3] K. D. Bierstedt and J. Bonet, *Some aspects of the modern theory of Frechet spaces*, RACSAM 97 (2003), 159–188.